



Arrays, Records and Pointers

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Outline

- Multidimensional Array
- Pointers: Pointer Array
- Records: Record Structure
- Representation of Records in Memory: Parallel Arrays
- Matrices
- Sparse Matrices

Multidimensional Array

- Two dimensional Array:

 $A_{J,K}$ or $A[J,K]$

		Columns			
		1	2	3	4
Rows	1	$A[1, 1]$	$A[1, 2]$	$A[1, 3]$	$A[1, 4]$
	2	$A[2, 1]$	$A[2, 2]$	$A[2, 3]$	$A[2, 4]$
	3	$A[3, 1]$	$A[3, 2]$	$A[3, 3]$	$A[3, 4]$

Fig. 4.8 *Two-Dimensional 3×4 Array A*

Multidimensional Array

- Representation of Two dimensional Array in memory:

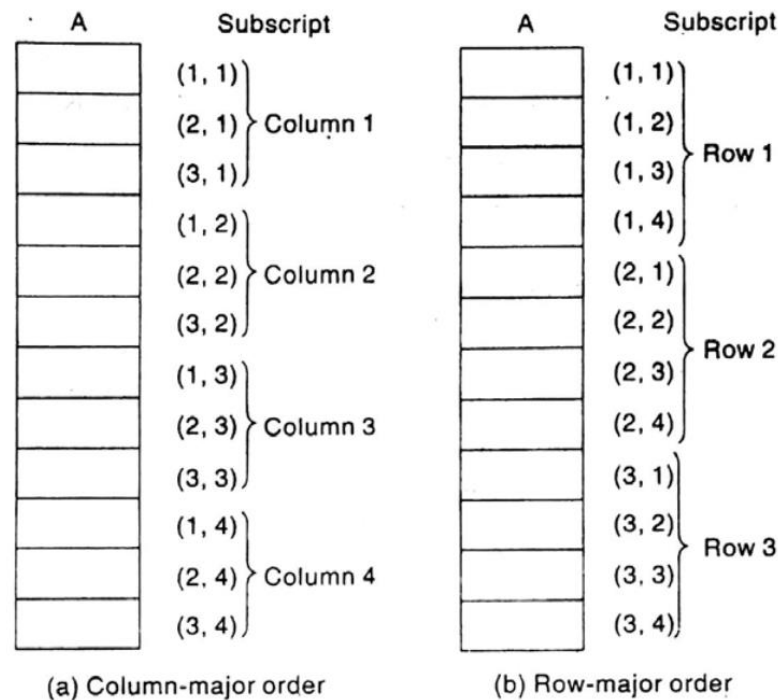


Fig. 4.10

Multidimensional Array

- General Multidimensional Arrays:

$B_{k_1, k_2, k_3, \dots, k_n}$ or $B[K_1, K_2, \dots, K_N]$

Multidimensional Array

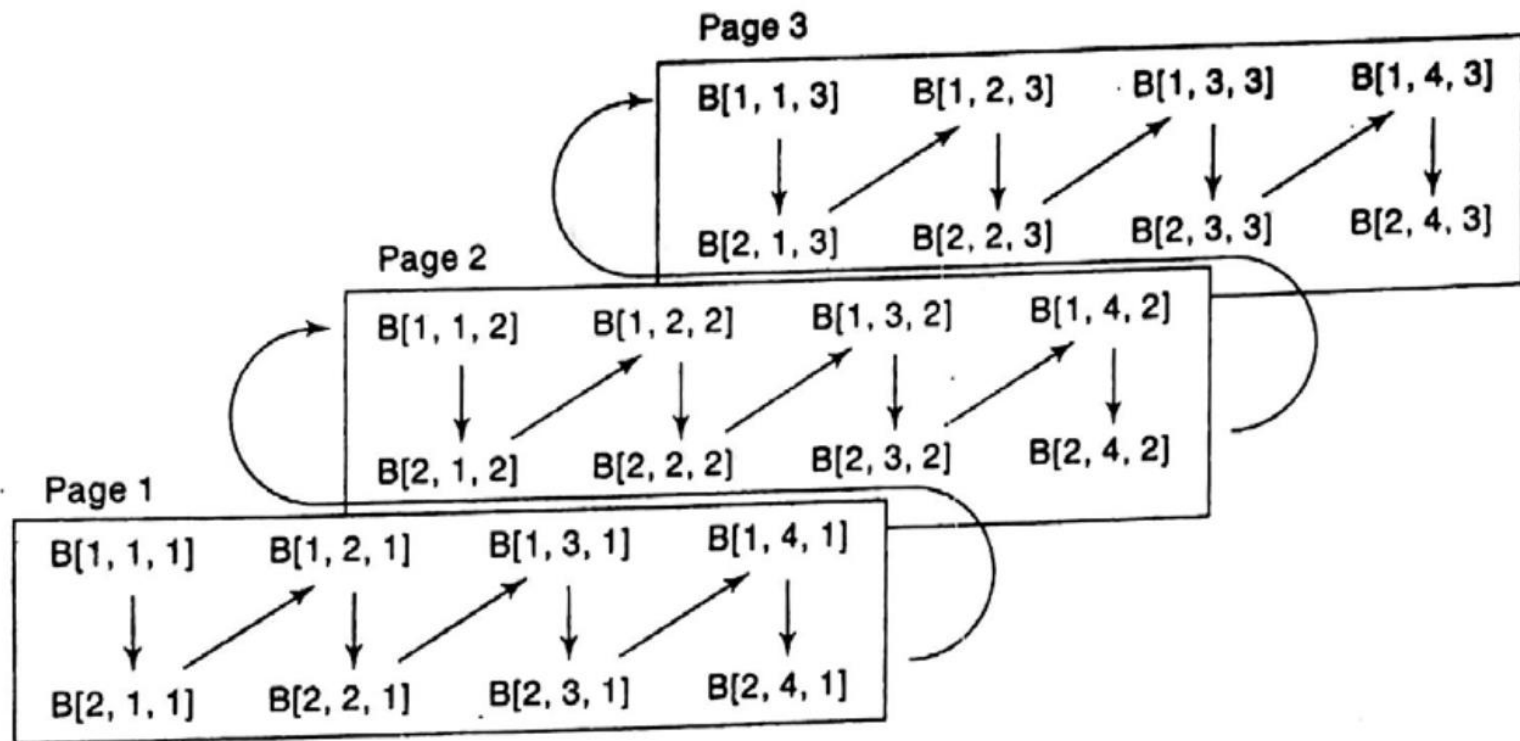


Fig. 4.11

Multidimensional Array

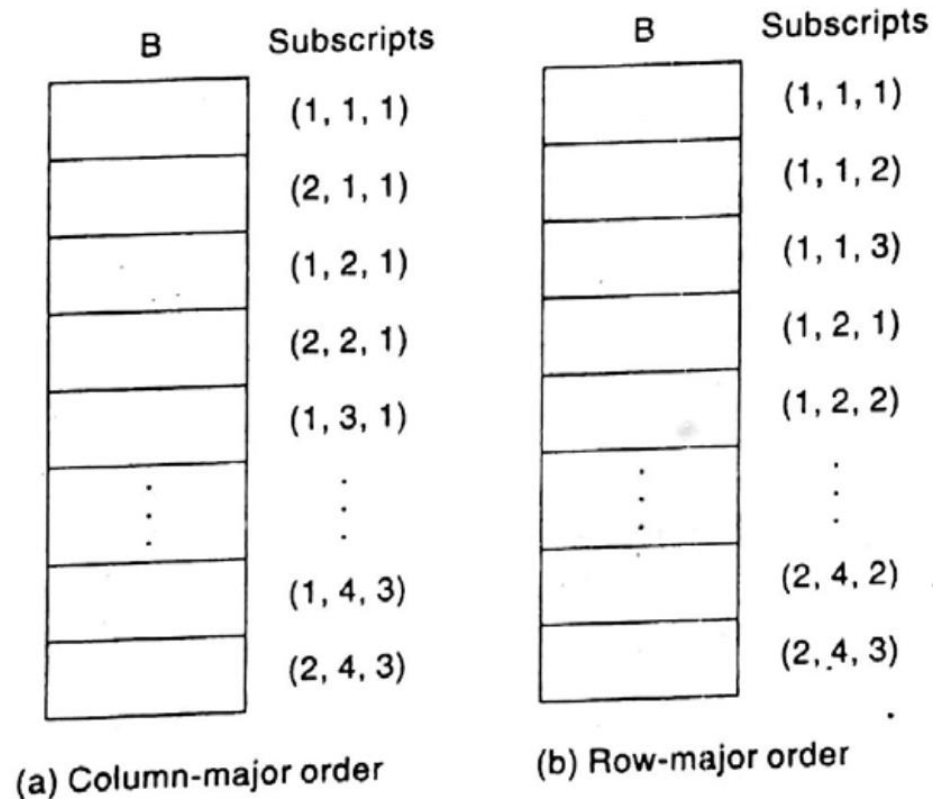


Fig. 4.12

Pointers: Pointer Array

Pointers: Pointer Array

- Pointer Array

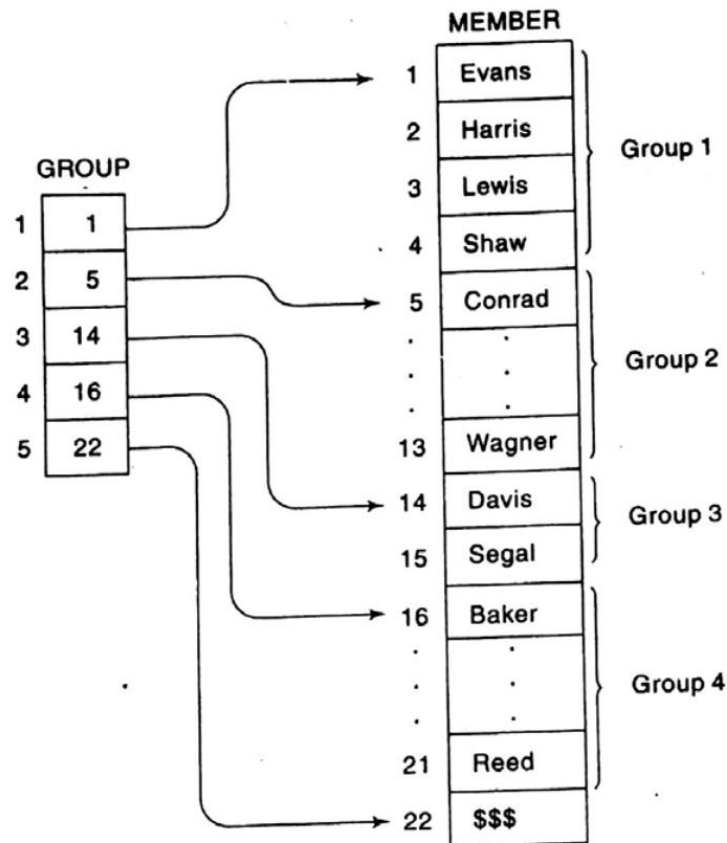


Fig. 4.16

Pointers: Pointer Array

- Pointer Array

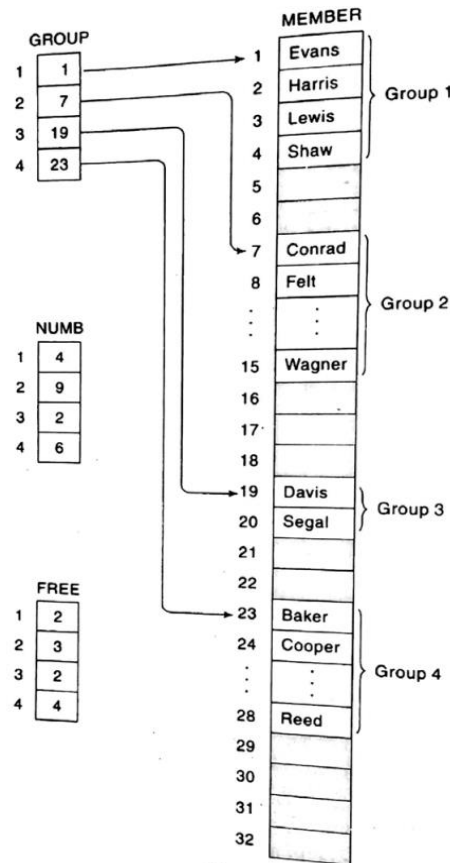


Fig. 4.17

Records: Record Structure

Records: Record Structure

- Differs from a linear array in the following ways –
 - A record may be a collection of nonhomogeneous data
 - The data items in a record are indexed by attribute name, so there may not be a natural ordering of its element.

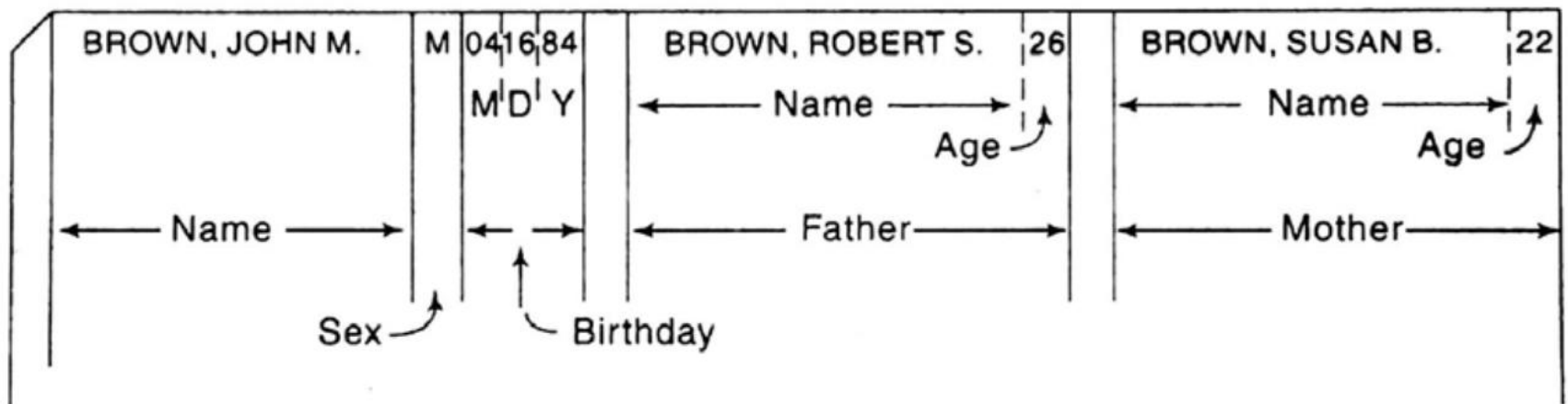


Fig. 4.18

Representation of Records in Memory: Parallel Arrays

Representation of Records in Memory: Parallel Arrays

	NAME	AGE	SEX	PHONE
1	John Brown	28	Male	234-5186
2	Paul Cohen	33	Male	456-7272
3	Mary Davis	24	Female	777-1212
4	Linda Evans	27	Female	876-4478
5	Mark Green	31	Male	255-7654
⋮	⋮	⋮	⋮	⋮

Fig. 4.19

Matrices

Matrices

- An n-elements **vector** V –

$$V = (V_1, V_2, \dots, V_n)$$

- An $m \times n$ **matrix** A –

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{pmatrix}$$

Matrices

- Scalar Product of two n-elements **vectors** V and U –

$$U \cdot V = U_1V_1 + U_2V_2 + \dots + U_nV_n = \sum_{k=1}^n U_kV_k$$

- Matrix multiplication

$$C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j} + \dots + A_{ip}B_{pj} = \sum_{k=1}^p A_{ik}B_{kj}$$

A is an $m \times p$

B is a $p \times n$ matrix.

Matrices

Algorithm 4.7: (Matrix Multiplication) MATMUL(A, B, C, M, P, N)
 Let A be an $M \times P$ matrix array, and let B be a $P \times N$ matrix array. This algorithm stores the product of A and B in an $M \times N$ matrix array C.

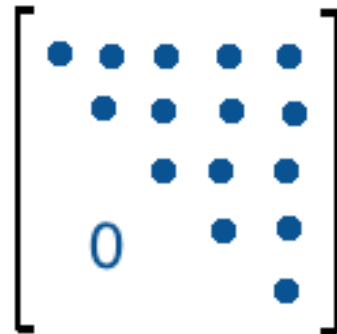
1. Repeat Steps 2 to 4 for $I = 1$ to M :
2. Repeat Steps 3 and 4 for $J = 1$ to N :
3. Set $C(I, J) := 0$. [Initializes $C(I, J)$.]
4. Repeat for $K = 1$ to P :
 $C(I, J) := C(I, J) + A[I, K] * B[K, J]$
 [End of inner loop.]
 [End of Step 2 middle loop.]
 [End of Step 1 outer loop.]
5. Exit.

Complexity: $O(n^3)$

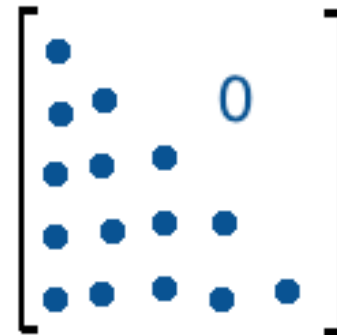
Sparse Matrices

Sparse Matrices

- High proportion of zero entries.



Upper Triangular
Matrix



Lower Triangular
Matrix

Sparse Matrices

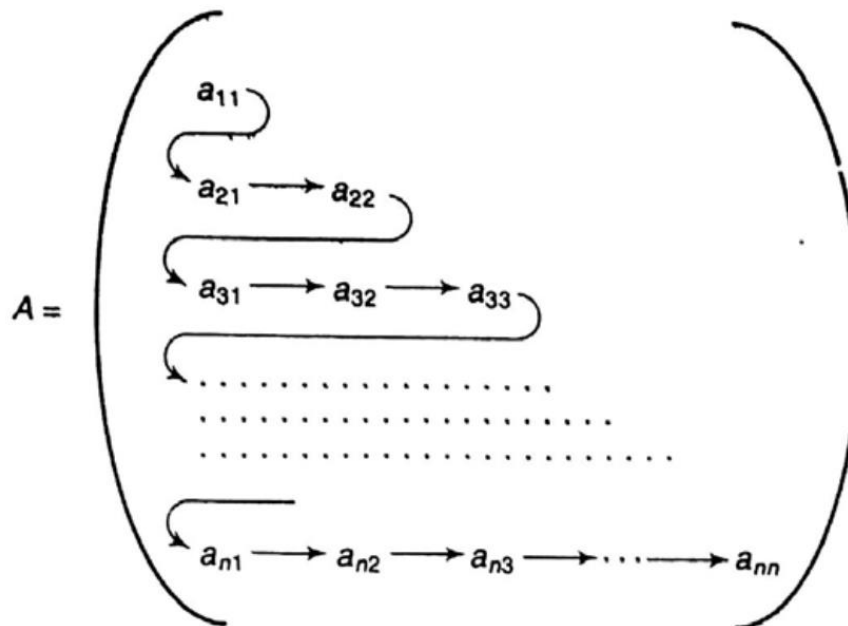


Fig. 4.22

$$B[1] = a_{11}, \quad B[2] = a_{21}, \quad B[3] = a_{22}, \quad B[4] = a_{31}, \quad \dots$$

Sparse Matrices

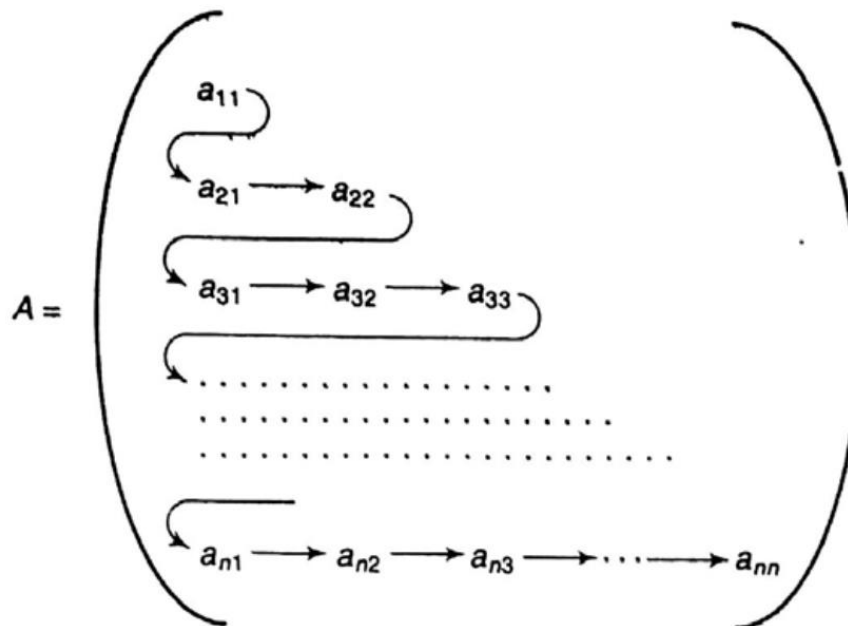


Fig. 4.22

$$1 + 2 + 3 + 4 + \dots + n = \frac{1}{2} n(n + 1)$$

Sparse Matrices

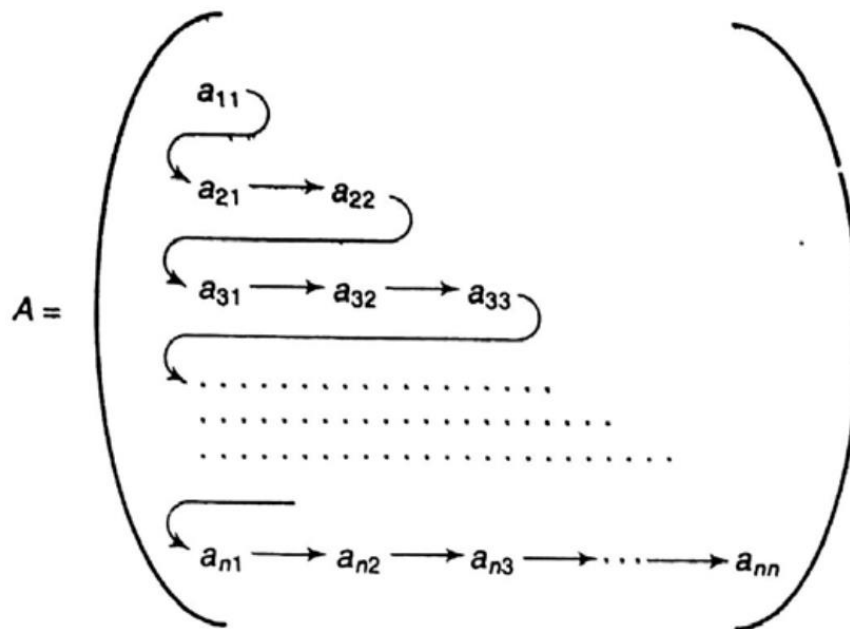


Fig. 4.22

1st row – 1 element
 2nd row – 2 elements

 Jth row – k elements

(J-1)th row –
 $1+2 + + (J-1)$

So $L = 1+2+... +(J-1) + K$

Sparse Matrices

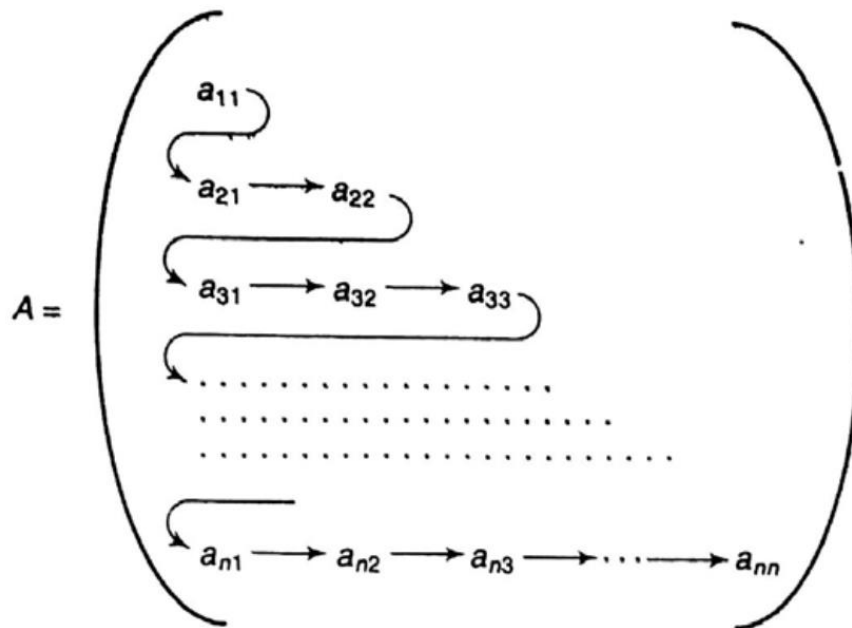


Fig. 4.22

$$B[L] = a_{JK}$$

$$L = \frac{J(J-1)}{2} + K$$

END